



In addition to Part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No.	: CE G525
Course Title	: Water Resources Planning and Management
Instructor-in-charge	: AJIT PRATAP SINGH
Website of I/C	: https://www.bits-pilani.ac.in/pilani/ajit-pratap-singh/ https://www.researchgate.net/profile/Ajit-Pratap-Singh

1. CE G525 Course Description

Introduction; Quantitative and qualitative assessment of water resources; Engineering principles applied to the management of water resources; Hydrographic and project surveys; Watershed management; Measurement techniques in water resources engineering; Gains of water resources planning to the society; Water economics; Computer utilization areas; Project discussions; Laboratory experiments.

2. Scope & Objective

The course *CE G525 Water Resources Planning and Management* aims to equip students with a comprehensive understanding of the engineering, economic, and environmental aspects involved in the sustainable planning and management of water resources. It covers both quantitative and qualitative assessments of water systems, integrating surface and groundwater hydrology, water quality modeling, and watershed management. The course emphasizes decision-making under uncertainty, project analysis, and the application of mathematical modeling and optimization techniques, including linear, non-linear, and multi-objective programming. Through the study of pollution sources, water conservation strategies, and demographic considerations, students gain insights into integrated water resource planning. Modern computational tools such as fuzzy logic and artificial intelligence are introduced alongside real-world case studies and laboratory experiments, enabling students to address complex, multidisciplinary challenges in water resource systems. The course ultimately prepares students to contribute effectively to socially and environmentally responsible water resource development and policy-making.

3. Course learning outcomes:

Upon successful completion of this course, students will be able to:

- Understand the structure, components, and modeling approaches of water resource systems.
- Apply principles of engineering and economics in water resource planning and project evaluation.
- Analyze point and non-point source pollution using water quality modeling techniques.
- Assess groundwater resources and apply methods for sustainable groundwater management.
- Perform statistical and uncertainty analyses related to streamflow and water availability.
- Formulate and solve water resource problems using linear, non-linear, and multi-objective optimization techniques.
- Plan and manage reservoir operations under both deterministic and stochastic environments.
- Utilize modern tools such as fuzzy logic and artificial intelligence in water resource management.
- Evaluate real-world case studies and apply integrated approaches to address complex water management challenges.

Text Books:

- T1. Vedula S. and P. P. Majumdar, Water Resources Systems: Modeling Techniques and Analysis, Tata McGraw Hill Publishing Company Limited, New Delhi, 2005.
- T2. Loucks D. P. and Eelco van Beek, Water Resources Systems Planning and Management: An Introduction to Methods, Models and Applications, UNESCO Publishing, 2005.



Reference Books/Journals

- R1. Goodman, Alvin S. Principles of Water Resources Planning, Prentice-Hall Inc, Englewood Cliffs, New Jersey, 1984.
 R2. Jain, S.K. and Singh V. P. (2003) Water Resources Systems Planning and Management, Elsevier.

Course Plan:

Week	Module Description	Lecture Number with Topic	Textbook Reference	Learning Outcome
1	Water Resource systems and role of modeling, planning and management: An Overview	L1: Introduction to Water Resource Systems L2: Scope and Need for Water Resources Planning L3: Role of Modeling in Water Resources Planning	T1: Ch.1, T2: Ch.1	Understand basic concepts, scope, and the importance of modeling in water resource systems.
2	Water Resource Systems: Its Role in Planning and Management	L4: System Approach and Water Resources Management L5: Project Analysis, Social Welfare Functions L6: Indifference Curves and Trade-offs	T1: Ch.2, T2: Ch.2	Learn principles of system modeling and project evaluation with socio-economic considerations.
3	Integrated water resource planning and conservation	L7: Data Needs for Planning – Demography & Demand L8: Water Conservation Techniques L9: Non-structural Alternatives in Conservation	T2: Ch.3, T1: Ch.2	Assess population-based water demands and conservation planning alternatives.
4	Models for Identifying and Evaluating Alternatives	L10: Evaluation of Alternative Designs and Policies L11: Environmental Impact Assessment L12: Integrated Water Resources Planning	T1: Ch.3, T2: Ch.4	Understand integrated approaches and environmental concerns in water resource projects.
5	Water Quality Modeling and Prediction	L13: Point and Non-Point Sources of Pollution L14: Water Quality Parameters and Standards L15: Water Quality Modeling – Mass Balance Approach L16: Water Quality Models – Plug Flow and Complete Mix	T2: Ch.5, T1: Ch.5	Analyze pollution sources and apply mass balance-based water quality modeling techniques. Apply analytical models for pollution control.
6	Groundwater resources and management	L17: Introduction to Groundwater Hydrology L18: Well Hydraulics and Yield Estimation	T1: Ch.6, T2: Ch.5	Understand the fundamentals of groundwater.
7	Groundwater resources and management	L19: Groundwater Management L20: Streamflow Data and Time Series Analysis L21: Frequency and Duration Analysis of Streamflow	T1: Ch.6 & 7, T2: Ch.6	Gain skills in groundwater planning and streamflow data analysis.



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Week	Module Description	Lecture Number with Topic	Textbook Reference	Learning Outcome
8	Decision making and uncertainty analysis	L22: Uncertainty in Water Resources Management L23: Introduction to Statistical Decision Theory L24: Risk Analysis in Planning	T1: Ch.8, T2: Ch.7	Analyze and incorporate uncertainty and risk in water resource decision-making.
9	Optimization Models and Methods in Water Resource Systems	L25: Introduction to Optimization Techniques L26: Linear Programming for Water Resources Problems L27: Non-Linear Programming Applications	T1: Ch.9, T2: Ch.8	Formulate and solve optimization models relevant to water system planning.
10	Reservoir Systems Management	L28: Reservoir Operation – Deterministic Models L29: Reservoir Operation – Stochastic Models L30: Simulation Techniques in Water Resources	T1: Ch.9, T2: Ch.8 & 9	Develop reservoir operation strategies using deterministic and stochastic frameworks.
11	Applications in Water Resources Planning and Management	L31: Case Study: Irrigation Planning Optimization L32: Case Study: Hydropower Optimization L33: Goal Programming in Water Resources	T1: Ch.10, T2: Ch.10	Apply optimization techniques to real-world water planning problems.
12	Applications in Water Resources Planning and Management	L34: Constraint and Weighting Techniques L35: Multi-Objective Decision Making L36: Introduction to Fuzzy Logic in Water Management	T1: Ch.10, T2: Ch.11	Understand multi-objective analysis and soft computing methods in water resource planning.
13	Applications in Water Resources Planning and Management	L37: Applications of Artificial Intelligence L38: Case Studies on Watershed and Reservoir Management L39: Case Studies on Pollution Control and Water Quality	T2: Ch.12, Supplementary Readings	Explore modern tools and evaluate real-life Indian case studies in water management.
14	Concluding Class	L40: Wrap-up, Discussion and Project Presentations, Review and Integration of All Modules	All chapters & supplementary resources	Integrate course knowledge and apply it in holistic project analysis and discussion.

Evaluation Scheme:

EC No.	Evaluation component	Duration	Weightage (%)	Date, Time and Venue	Nature of component
1	Mid semester	90 min.	25	05/10 AN2	OB
2	Project* (Includes term paper and seminar)		30	To be announced from time to time	OB



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	Mid-Semester Project Presentation and Report	10-15 min for each Project	To be announced	Last week of October	OB
	Final Project Presentation and Report	15-25 min for each Project	To be announced	First week of December	OB
3	Class Participation and Home Assignments		10	To be announced from time to time	OB (Take Home)
4	Comprehensive Examination	3 Hours	35	03/12 (AN)	CB

*During the semester, each student will work in a team on a project dealing with water resources planning and management. These projects will deal with some aspect of a real, complex water resources planning and analysis issue of current interest to the world, nation, or a particular State. The projects typically are interdisciplinary and consider such aspects of water resources planning and management as: engineering, law; economics; hydrology, climatology, hydrogeology; environmental health; computer modeling; population growth; land use planning; regional development; or politics. Each group will make an oral presentation of their progress and outcome to the class during Mid-semester presentation and final presentation and submit the reports accordingly.

The purposes of the term project are:

1. To enable you to explore in-depth an aspect of the subject of water resources planning and management.
2. To provide experience in the formulation, execution and presentation of an engineering investigation in the area of water resources planning and management.
3. To provide a teamwork experience dealing with the interdisciplinary nature of water resources planning and management.

Steps in Carrying out the Project

The steps in carrying out the project are:

1. Prepare a proposal in MS Word and email it to the instructor specifying the objective of your project and outlining how you plan to go about executing it. Before beginning your project, you need to prepare a project proposal and submit it to the instructor for approval and feedback. This proposal should be about 1-2 page in length. The instructor will review the proposals and provide comments for the student to revise the proposal for final submittal and approval. The proposal should contain:
 - Title of the project,
 - Team members.
 - Scope, Definition of the problem and Background of the study,
 - Objectives,
 - Plan work,
 - Discussion (preliminary) of the proposed methods of solution of the problem.
 - Anticipated data needs.
 - Anticipated problems in carrying out the project.
 - Assignment of tasks to team members.
 - Anticipated outcome
 - A minimum of 5 bibliographic/references citations relevant to your proposed project.
2. Present an oral report in class on the date posted on the class assignment web page.
3. Send a written report (in MS Word) to the instructor on the date posted on the class assignment web page.

Chamber Consultation Hour: Tuesday 4-5 pm in room no. 2112 or through Google Meet.

Notices: All notices concerning the course will be displayed on Civil Engineering Department Notice Board/ Google Class Room. Also, classes shall be engaged online through Google Meet.

Make-up Policy:

1. Make-up will be granted only on genuine reasons. However, prior permission is must.
2. For medical cases, a certificate from the concerned physician of the Medical Centre must be produced.



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Minimum Clearing/Passing Score Policy of the Course: A student must obtain 30% of the average of the top two performers in the class, or 40% of the median of the class, whichever is lower to clear the course. This shall be considered as the minimum performance criterion to clear/pass the course. If any student gets the marks lower than the prescribed standard mentioned above, he/she will be awarded NC.

Attendance Clause

As BITS Pilani is a fully residential university with an internal and continuous evaluation system, it is expected that every student demonstrates a high level of responsibility by maintaining 100% attendance in all classes and laboratory sessions. Regular attendance and class participation are integral components of the learning process, and students are expected to appear in all scheduled tests and examinations and complete assigned academic tasks sincerely and on time.

While 100% attendance is the expected norm, under no circumstances should a student's attendance fall below 60% in this course. A student falling short of this minimum attendance requirement shall not be permitted to appear in the comprehensive examination and will be considered as not having cleared the course.

Students are therefore advised to remain fully aware of their academic responsibilities and ensure consistent attendance throughout the semester.

Academic honesty and integrity Policy:

Academic honesty and integrity of high quality is expected to be maintained by all of the students throughout the Semester and any kind of academic dishonesty or indulgence in illegitimate practices is not acceptable.

Instructor-in-charge
CE G525



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FIRST SEMESTER, 2026-2027

(Outline of the course: To be filled in consultation with the proposed supervisor*)

Course No.: CE G525 Course Title: Water Resources Planning and management

Name of the Instructor/Supervisor: Prof. Ajit Pratap Singh

Title of the Course Project work:

Team members (Name of the students & ID Nos.):

Scope, Definition of the problem and Background of the study,

Objectives of the Study:

Plan of Work:

Discussion (preliminary) of the proposed methods of solution of the problem.

Anticipated data needs.

Anticipated problems in carrying out the project.

Assignment of tasks to team members.

Anticipated outcome

A minimum of 5 bibliographic/references citations relevant to your proposed project.

Student's Signature:

Approved/Disapproved

Date:

Signature of the instructor

***The registered student must submit duly filled above outline after verification from the I/C to the Instructor-in-charge of the course, Prof. A. P. Singh latest by 24.08.2026.**



Guidelines for the Award of Punishment to Unfair Means Cases (On Campus Programs)

Code	Offence	Punishment
1.0	Possession of any communication device such as a mobile phone in the examination hall; possession of any written material generally not allowed during examinations.	Registration cancelled for the said course
2.0	Possession of material, in any form, relevant to the subject but not caught in the act of copying.	Registration cancelled for the said course
3.0	Simple* cases of copying during the examinations /Simple* cases of tampering with the answer book either during the examination or during the distribution of answer scripts	Registration cancelled for the said course
4.0	Severe** cases of copying during the examination / Severe** cases of tampering with the answer book either during the examination or during the distribution of answer scripts	Registration cancelled for the said course and debarred to register for one semester to two years, including summer terms, depending on the degree of severity of the offence.
5.0	Impersonation (deputing a person on one's behalf to write the examination or appearing for a person to write the examination)	Registration cancelled for the said course and discontinued from the programme

* Simple cases of copying/tampering includes spontaneous/ non-premeditated copying or tampering.

** Severe cases of unfair means includes premeditated copying (such as copying from material brought into the examination hall with the intension of copying therefrom, including material stored on cell phone, in print form etc.), copying in collaboration with other student(s), helping a student in the examination hall by unfair means, premeditated tampering etc. Degree of severity to be decided by the examination sub-committee in respective campuses.

Note:

1. The word examination includes any component of evaluation in the course.
2. In case of second and further offences committed by a student, sub-committee of respective campuses will take suitable decision depending on the degree of severity of the offence.
3. For any other offence not covered above, the Examination Sub Committee can take appropriate action and report to the Examination Committee.
4. In case of any appeal, the Sub Committees of Examination Committee will go through the appeals and send their recommendations along with the appeal to the Examination Committee for the final decision.